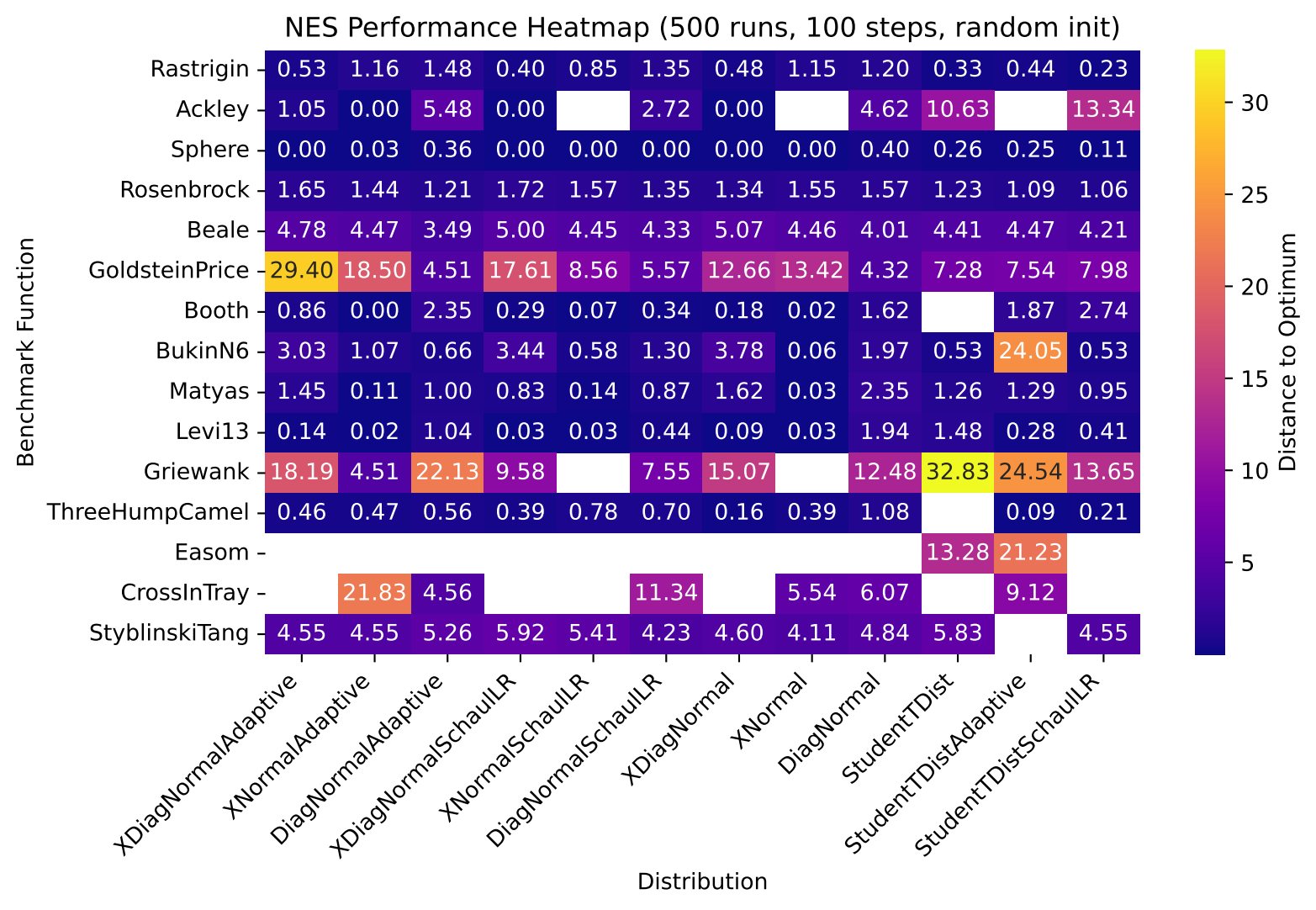
NES Performance Heatmap (500 runs, 100 steps, random init)
Benchmark Function
Rastrigin 0.53 1.16 1.48 0.40 0.85 1.35 0.48 1.15 1.20 0.33 0.44 0.23
Ackley 1.05 0.00 5.48 0.00 2.72 0.00 4.62 10.63 13.34
Sphere 0.00 0.03 0.36 0.00 0.00 0.00 0.00 0.00 0.40 0.26 0.25 0.11
Rosenbrock 1.65 1.44 1.21 1.72 1.57 1.35 1.34 1.55 1.57 1.23 1.09 1.06
Beale 4.78 4.47 3.49 5.00 4.45 4.33 5.07 4.46 4.01 4.41 4.47 4.21
GoldsteinPrice 29.40 18.50 4.51 17.61 8.56 5.57 12.66 13.42 4.32 7.28 7.54 7.98
Booth 0.86 0.00 2.35 0.29 0.07 0.34 0.18 0.02 1.62 1.87 2.74
BukinN6 3.03 1.07 0.66 3.44 0.58 1.30 3.78 0.06 1.97 0.53 24.05 0.53
Matyas 1.45 0.11 1.00 0.83 0.14 0.87 1.62 0.03 2.35 1.26 1.29 0.95
Levi13 0.14 0.02 1.04 0.03 0.03 0.44 0.09 0.03 1.94 1.48 0.28 0.41
Griewank 18.19 4.51 22.13 9.58 7.55 15.07 12.48 32.83 24.54 13.65
ThreeHumpCamel 0.46 0.47 0.56 0.39 0.78 0.70 0.16 0.39 1.08 0.09 0.21
Easom 13.28 21.23
CrossInTray 21.83 4.56 11.34 5.54 6.07 9.12
StyblinskiTang 4.55 4.55 5.26 5.92 5.41 4.23 4.60 4.11 4.84 5.83 4.55
XDiagNormalAdaptive
XNormalAdaptive
DiagNormalAdaptive
XDiagNormalSchaulLR
XNormalSchaulLR
DiagNormalSchaulLR
XDiagNormal
XNormal
DiagNormal
StudentTDist
StudentTDistAdaptive
StudentTDistSchaulLR
Distribution
Distance to Optimum
30
25
20
15
10
5